

NOOK

Property Rental Platform

Installation & Setup Guide

Version	Date
1.0.0	April 2026

PFP-CP2 Team — ENSTA
Namane Haithem, Akli Merouane, Bennacer Sami Fares,
Tetbirt Abdellah, Baroudi Laid, Djabed Alae dine, Messikh Yahia

1. Introduction

This guide provides step-by-step instructions to install and run the Nook application on a local development machine. Nook is a full-stack web application for property rentals built with a Django backend and a Next.js frontend.

Target audience: developers and technical administrators who want to set up the project locally.

2. Prerequisites

Ensure the following software is installed on your machine before proceeding:

Software	Min. Version	Notes
Node.js	v18+	Required for the Next.js frontend
npm	v9+	Bundled with Node.js
Python	v3.10+	Required for the Django backend
pip	latest	Python package manager
PostgreSQL	v14+	Project uses Supabase (cloud) by default
Redis	v6+ (optional)	Used for caching; skip for basic setup
Git	latest	To clone the repository

3. Getting the Project Files

If you received the project as a compressed archive (.zip), follow these steps first:

3.1 Unzip the Archive

Locate the downloaded file (e.g., PFP-CP2.zip) and extract it:

```
# On Windows (PowerShell):
Expand-Archive -Path PFP-CP2.zip -DestinationPath C:\Projects\PFP-CP2

# On macOS / Linux:
unzip PFP-CP2.zip -d ~/Projects/PFP-CP2
```

After extraction, navigate into the project root:

```
cd PFP-CP2
```

You should see the following top-level structure:

```
PFP-CP2/
├── backEnd/      # Django backend
├── frontEnd/
│   └── pfp_front/ # Next.js frontend
```

4. Backend Setup (Django)

4.1 Navigate to the Backend Directory

```
cd backEnd
```

4.2 Create and Activate a Virtual Environment

```
# Windows
python -m venv venv
venv\Scripts\activate

# macOS / Linux
python3 -m venv venv
source venv/bin/activate
```

4.3 Install Python Dependencies

```
pip install -r requirements.txt
```

4.4 Configure Environment Variables

Create a `.env` file in the `backEnd/` directory and fill in the values below:

```
DEBUG=True
SECRET_KEY=your-django-secret-key
DATABASE_URL=postgres://user:password@host:port/dbname
REDIS_URL=redis://127.0.0.1:6379/1
AWS_ACCESS_KEY_ID=your-s3-access-key
AWS_SECRET_ACCESS_KEY=your-s3-secret-key
```

Note — Supabase

The project is pre-configured to use a Supabase PostgreSQL instance and Supabase Storage (S3-compatible). You may use the provided credentials or point to your own database. Redis is optional for basic local development.

4.5 Run Database Migrations

```
python manage.py migrate
```

4.6 Start the Backend Server

```
python manage.py runserver
```

The API will be available at: `http://127.0.0.1:8000/`

Interactive API docs (Swagger UI) will be available at: `http://127.0.0.1:8000/api/docs`

5. Frontend Setup (Next.js)

5.1 Navigate to the Frontend Directory

Open a new terminal window and run:

```
cd frontEnd/pfp_front
```

5.2 Install Node.js Dependencies

```
npm install
```

5.3 Configure Environment Variables

Create a .env.local file in the frontEnd/pfp_front/ directory:

```
NEXT_PUBLIC_API_URL=http://127.0.0.1:8000/api
```

5.4 Start the Development Server

```
npm run dev
```

The frontend application will be available at: <http://localhost:3000/>

6. Verifying the Installation

Once both servers are running, perform these quick checks to confirm the setup is working:

#	Check	URL	Expected Result
1	Frontend home page	http://localhost:3000	Nook home page loads
2	Backend API root	http://127.0.0.1:8000/api/docs	Swagger UI appears
3	Login endpoint	POST /api/token/	Returns JWT tokens
4	Properties listing	GET /api/Posts/	Returns JSON list

7. Troubleshooting

Below are the most common errors and their solutions:

7.1 ModuleNotFoundError (Python)

```
ModuleNotFoundError: No module named 'ninja'
```

Solution: Ensure your virtual environment is activated and re-run:

```
pip install -r requirements.txt
```

7.2 Database Connection Error

```
django.db.utils.OperationalError: could not connect to server
```

Solution: Verify the DATABASE_URL in your .env file matches your PostgreSQL credentials. If using the default Supabase instance, check your internet connection.

7.3 npm install Fails

```
npm ERR! peer dep missing
```

Solution: Delete the node_modules folder and the package-lock.json file, then reinstall:

```
rm -rf node_modules package-lock.json  
npm install
```

7.4 Port Already in Use

```
Error: address already in use :::3000
```

Solution: Kill the process occupying the port or run on an alternate port:

```
# Kill the process (Linux/macOS)  
kill -9 $(lsof -t -i:3000)  
  
# Or start frontend on a different port  
npm run dev -- --port 3001
```

Redis Not Running

If Redis is not installed or not running, the backend will still function but caching will be disabled. You may see a connection warning in the terminal — this is non-critical for local development.